obeys the exponential decay law, we can also write an equation for an activity similar to equation (9.36).

After n half-lives, the activity or decay rate of any radioactive sample is

$$R = \left(\frac{1}{2}\right)^n R_0 \qquad (9.41)$$

Mean life (τ):

When the radioactive nucleus decays, the nucleus which disintegrates first has zero life time and the nucleus which decays last has an infinite lifetime. The actual life time for each nucleus varies from zero to infinity. Therefore, it is meaningful to define average life or mean life time θ, that the nucleus survives before it decays.

The mean life time of the nucleus is the ratio of sum or integration of life times of all nuclei to the total number nuclei present initially.

The total number of nuclei decaying in the time interval from t to $t+\Delta t$ is equal to $R\Delta t = \lambda N_0 e^{-\lambda t}\Delta t$. It implies that until the time t, this $R\Delta t$ number of nuclei lived. So the life time of these $R\Delta t$ nuclei is equal to $tR\Delta t$. In the limit $\Delta t \to 0$, the total life time of all the nuclei would be the integration of $tRdt$ from the limit $t=0$ to $t=\infty$.

Mean life

$$\tau = \frac{\int_0^\infty t[Rdt]}{N_0} = \frac{\int_0^\infty t\left[\lambda N_0 e^{-\lambda t}\, dt\right]}{N_0} \qquad (9.42)$$

After a few integration (refer box item), the expression for mean life time,

$$\tau = \frac{1}{\lambda} \qquad (9.43)$$

Note that mean life and decay constant are inversely proportional to each other.

Using mean life, the half-life can be rewritten as

$$T_{1/2} = \tau \ln 2 = 0.6931\tau \qquad (9.44)$$

Mean life : Not for examination

The integration in the equation (9.42) can be performed using integration by parts.

$$\tau = \frac{\int_0^\infty \lambda N_0 t\, e^{-\lambda t} dt}{N_0} = \frac{\lambda N_0 \int_0^\infty t\, e^{-\lambda t} dt}{N_0}$$

$$\tau = \lambda \int_0^\infty t\, e^{-\lambda t} dt$$

$$u = t \qquad dv = e^{-\lambda t}\, dt$$

$$\tau = \lambda \int_0^\infty t\, e^{-\lambda t} dt = \lambda \left[\frac{t\, e^{-\lambda t}}{-\lambda}\right]_0^\infty - \lambda \int_0^\infty \left[\frac{e^{-\lambda t}}{-\lambda}\right] dt$$

By substituting the limits, the first term in the above equation becomes zero.

$$\tau = \int_0^\infty e^{-\lambda t}\, dt = -\frac{1}{\lambda}\left[e^{-\lambda t}\right]_0^\infty = \frac{1}{\lambda}$$

EXAMPLE 9.12

Calculate the number of nuclei of carbon-14 undecayed after 22,920 years if the initial number of carbon-14 atoms is 10,000. The half-life of carbon-14 is 5730 years.

Solution

To get the time interval in terms of half-life,

$$n = \frac{t}{T_{1/2}} = \frac{22{,}920\, yr}{5730\, yr} = 4$$

The number of nuclei remaining undecayed after 22,920 years,

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